# Day 12 – Linux Networking (Theory Assignment)

Recommended file name: `StudentName_Day12_LinuxNetworking.pdf`

Below are all 20 answers in simple, student-friendly language. You can use this content for your assignment and GitHub repository.

## THEORY – 20 Questions and Answers

1. What is a network interface?

A network interface is a hardware or software component that allows a computer to connect and communicate with a network. Examples include Ethernet and Wi-Fi interfaces.

2. What is the loopback interface?

The loopback interface is a special network interface that allows a computer to communicate with itself. In Linux, it is commonly named `lo` and uses the IP address `127.0.0.1`.

3. What does `ip addr` display?

The `ip addr` command displays information about network interfaces, including their IP addresses, interface names, and network status.

4. What is the purpose of `ip link`?

The `ip link` command displays and manages network interface information. It can be used to check whether an interface is active or inactive.

5. What does `hostname -I` display?

The `hostname -I` command displays the IP addresses assigned to the system's network interfaces.

6. What does `ip route show` display?

The `ip route show` command displays the routing table, which contains information about how network traffic is sent to different destinations.

7. What is a default route?

A default route is the network path used when a computer does not have a specific route for a destination. It usually directs traffic to the network's default gateway.

8. What does `ping` test?

The `ping` command tests network connectivity between two devices by sending ICMP echo requests and checking for replies. It also provides information about response time and packet loss.

9. Why can a failed ping be misleading?

A failed ping does not always mean that a device is offline. Firewalls, security settings, or network configurations may block ICMP traffic even when the device is working properly.

10. What is DNS?

DNS stands for Domain Name System. It translates human-readable domain names, such as `google.com`, into IP addresses that computers use to communicate.

11. What does `getent hosts` do?

The `getent hosts` command looks up host information using the system's configured name resolution services. It can return the IP address associated with a hostname.

12. What information does `ip neigh` display?

The `ip neigh` command displays information about neighboring devices on a local network. It shows IP addresses, associated MAC addresses, and neighbor connection states.

13. What is a socket?

A socket is a communication endpoint that allows applications to send and receive data over a network. It is commonly identified by an IP address and port number.

14. What does `ss -tuln` display?

The `ss -tuln` command displays listening network sockets.

- `-t`: Shows TCP sockets.
- `-u`: Shows UDP sockets.
- `-l`: Shows listening sockets.
- `-n`: Displays numerical addresses and port numbers.

15. What is the difference between TCP and UDP conceptually?

TCP and UDP are communication protocols used to transfer data over networks.

- TCP: Provides reliable, connection-oriented communication. It uses acknowledgements and retransmission to help ensure data delivery.
- UDP: Provides connectionless communication with lower overhead. It does not guarantee delivery or packet order.

Learn more

16. What is curl used for?

`curl` is a command-line tool used to transfer data to and from servers using protocols such as HTTP and HTTPS. It is useful for testing web services and checking server responses.

17. What does `curl -I` request?

The `curl -I` command sends an HTTP HEAD request to a server and displays the response headers without downloading the response body.

18. What is traceroute used for?

Traceroute is a network diagnostic tool used to identify the path that packets take from one computer to another. It helps identify network delays and possible routing problems.

19. Why is ss commonly preferred over netstat on modern Linux?

`ss` is commonly preferred because it is faster, provides detailed socket information, and is part of the modern Linux networking tools. The older `netstat` utility belongs to the deprecated net-tools package.

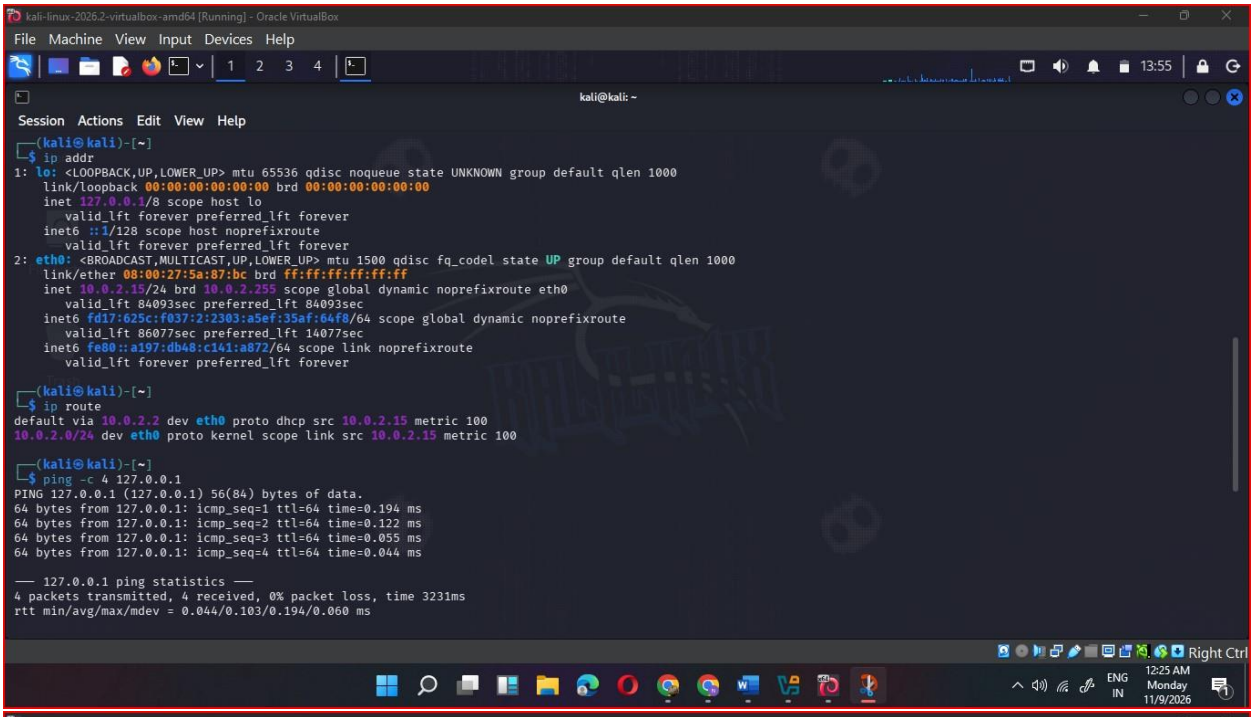
20. Why are Linux networking commands useful in cybersecurity?

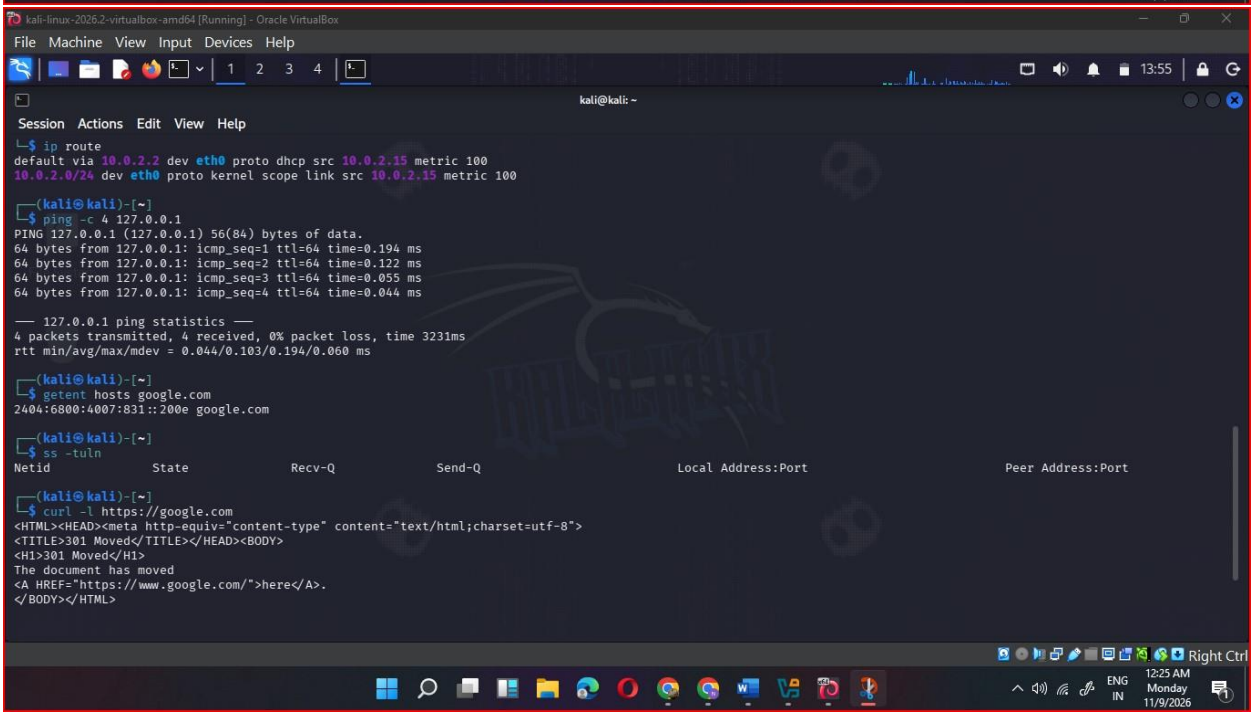
Linux networking commands are useful in cybersecurity because they help security professionals understand network configurations, check connectivity, identify listening services, troubleshoot network problems, and monitor network activity during authorized security testing.

# Basic Linux Networking Commands

| Command | Purpose |
|---|---|
| `ip addr` | Displays IP addresses |
| `ip link` | Displays network interfaces |
| `hostname -I` | Displays assigned IP addresses |
| `ip route show` | Displays routing information |
| `ping google.com` | Tests network connectivity |
| `getent hosts google.com` | Resolves hostname information |
| `ip neigh` | Displays neighboring devices |

| | |
|---|---|
| `ss -tuln` | Displays listening TCP/UDP sockets |
| `curl -I https://example.com` | Displays HTTP response headers |
| `traceroute google.com` | Displays the network path |


kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: ~
Session Actions Edit View Help
(kali㉿kali)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
valid_lft 84093sec preferred_lft 84093sec
inet6 fd17:625c:f037:2:2303:a5ef:35af:64f8/64 scope global dynamic noprefixroute
valid_lft 86077sec preferred_lft 14077sec
inet6 fe80::a197:db48:c141:a872/64 scope link noprefixroute
valid_lft forever preferred_lft forever
(kali㉿kali)-[~]
$ ip route
default via 10.0.2.2 dev eth0 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.15 metric 100
(kali㉿kali)-[~]
$ ping -c 4 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.194 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.122 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.055 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.044 ms
— 127.0.0.1 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3231ms
rtt min/avg/max/mdev = 0.044/0.103/0.194/0.060 ms
Right Ctrl
ENG IN
12:25 AM Monday 11/9/2026


kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: ~
Session Actions Edit View Help
$ ip route
default via 10.0.2.2 dev eth0 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.15 metric 100
(kali㉿kali)-[~]
$ ping -c 4 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.194 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.122 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.055 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.044 ms
— 127.0.0.1 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3231ms
rtt min/avg/max/mdev = 0.044/0.103/0.194/0.060 ms
(kali㉿kali)-[~]
$ getent hosts google.com
2404:6800:4007:831::200e google.com
(kali㉿kali)-[~]
$ ss -tuln
Netid State Recv-Q Send-Q Local Address:Port Peer Address:Port
(kali㉿kali)-[~]
$ curl -l https://google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html;charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="https://www.google.com/">here</A>.
</BODY></HTML>
Right Ctrl
ENG IN
12:25 AM Monday 11/9/2026